

03-12-2025

↳ picking up from 2/24

$$V^R(k, x, t) = \max_{c, k'} \{ u(c) + \beta E V^R(k', x', t+1) \}$$

↳ = ufunc

$$\sum_{x_{t+1}} \Psi(x_{t+1} | x_t) V^R(k_p, x_{t+1}, t+1)$$

Eval

function u = ufunc(kp, t, ix, ik)

global VR xgrid kgrid r β ψ ss  
 cons = kgrid(ik)(1+r) + ss - xgrid(ix) - kp

utiltoday = ln(cons)

locate kp on kgrid ("find" command)

weight = kp - kgrid(jlo) / kgrid(jlo+1) - kgrid(jlo)

bounds

no x

↳ Vint = (1-weight) \* VR(jlo, t+1) + weight \* VR(jlo+1, t+1)

w/x:

Eval = 0

for ixp = 1:m

Vint = (1-weight) \* VR(jlo, ixp, t+1) + weight \* VR(jlo+1, ixp, t+1)

Eval = Eval + ψ(ixp | ix) \* Vint

end

$$U = u_{today} + \beta \cdot eval$$

$$\star U = \underline{\underline{-U}}$$

→ Back to main

$[fval, kval] = fminsearch(@kp ufunc(kp, t, ix, ik))$

$VR(ik, ix, t) = -fval$

$optkr = kval$

$totres(ik, ix, t) = totres - kval$

end

end

end

for  $t = R-1:1$        $Vtemp(1:2)$   
                           $Ctemp(1:2)$

for  $iz = 1:iv$        $ktemp(1:2)$

for  $ix = 1:m$

for  $ik = 1:n$

for  $il = 1:2$  ← labor choice; 1: no work, 2: work

$Totres = kgrid(ik)(1+r) - xgrid(ix) + zgrid(iz) \cdot w \cdot (il-1)$

if  $totres < cmin$

⋮

else

$Ufuncw(kp, t, ik, ix, iz, il)$

global  $(VR, kgrid, xgrid, zgrid, r, \beta, \psi, SS, \phi)$

$$\text{cons} = k_{\text{grid}}(ik) + lab_{\text{inc}} - x_{\text{grid}}(ix) - k_p$$

if  $il = 1$

$lab_{\text{inc}} = 0, dis = 0$

else if  $il = 2$

$lab_{\text{inc}} = w \cdot z_{\text{grid}}(iz), dis_{\text{util}} = \emptyset$

$$U_{\text{today}} = \ln(\text{cons}) - dis_{\text{util}}$$

Same process for weight + interp

$$U(c, l) + \beta E V^w(k', z', x', t+1)$$

$$\sum_{x'} \sum_{z'} \Psi(x_{t+1} | x_t) \Pi(z_{t+1} | z_t) V(k', x', z', t')$$

$$Eval = 0$$

for  $ixp = 1:m$

    for  $izp = 1:v$

$$\begin{aligned} V_{\text{int}} &= (1 - \text{weight}) V(jlo, ixp, izp, t+1) \\ &\quad + \text{weight} V(jlo+1, ixp, izp, t+1) \end{aligned}$$

$$Eval = Eval + \Psi \cdot T \cdot V_{\text{int}}$$

end

end

$$U = U_{\text{tilt today}} + \beta E_{\text{val}}$$

$$U = -U$$

Back to main:

$$[f_{\text{val}}, k_{\text{val}}] = \dots$$

$$V_T(l_{il}) = -f_{\text{val}}$$

$$k_T(l_{il}) = k_{\text{val}}$$

end loop  $il$

Compare  $V_T(1) + V_T(2)$

i.e.

$$\text{if } V_T(1) > V_T(2)$$

$$V(ik, ix, iz, t) = V_T(1)$$

$$\text{opt}k(ik, ix, iz, t) = k_T(1)$$

$$\text{opt}l(ik, ix, iz, t) = 1$$

else

$$V(ik, ix, iz, t) = V_T(2)$$

$$\text{opt}k(ik, ix, iz, t) = k_T(2)$$

$$\text{opt}l(ik, ix, iz, t) = 2$$

end

end

end

end

Simulation:  $\begin{matrix} \nearrow z_{sim} \\ x_{sim}(1:T), z(1:R-1) \Rightarrow \text{random} \end{matrix}$

Put  $k(1)$  on grid

Start from  $t=1 \Rightarrow k_1=0$

$$k_{sim}(1) = \text{opt}k(1, x_{sim}(1), z_{sim}(1), 1)$$

$t=2$ :

locate  $k_{sim}(1) \Rightarrow \text{weight}$

$$k_{sim}(2) = (1 - \text{weight}) \cdot \text{opt}k(j_0, x_{sim}(2), z_{sim}(2), 2) \\ + \text{weight} \cdot \text{opt}k(j_0+1, x_{sim}(2), z_{sim}(2), 2)$$